

Integrating Vision and Multimodal Language Models for Collaborative Building Footprint Quality Assessment

Extracting building footprints has been an important topic in remote sensing for many years. Most methods are based on deep learning and convolutional neural networks (CNNs), which are trained on large datasets for specific use cases (Rastogi et al., 2022; O'Shea & Nash, 2015). These models have been used in many fields, such as disaster management, urban planning, and infrastructure monitoring (Awrangjeb et al., 2010). However, CNNs are limited because they need large labeled data and do not generalize well to new areas.

In recent years, with the rise of artificial intelligence, researchers have started to use foundation models instead of classical CNNs. The Dense Prediction Transformer (DPT) by (Ranftl et al., 2021) was one of the first to show how transformer-based models can replace CNNs. These new models can look at the whole image at once and make more coherent predictions.

Today, research is moving toward combining visual foundation models with multimodal large language models (MLLMs) (Wu & Gan, 2023). These models can understand both text and images and are being studied for how well they can process spatial data (Ji et al., 2025) and interact with vision model. In remote sensing, first studies have started to combine these two types of models (Zhou et al., 2024), usually so that the language model helps describe or guide what the visual model should segment.

My project takes a different approach. Instead of using the language model to control the vision model, I want to generate rough building polygons with a visual foundation model and then use a multimodal language model to check and evaluate them. The language model will identify possible errors and suggest how to improve the polygons. This is my research gap combining these two model types in a new way. Although the focus is on building footprints, the same idea could also be used for other remote sensing quality checks.

Rough Research questions:

- **RQ1:** How accurately can MLLMs detect and describe typical geometric errors in building polygons?
- **RQ2:** How can the feedback from an MLLM be converted into spatial prompts to refine segmentation results?
- **RQ3:** How does the combined pipeline perform compared to conventional CNN-based approaches? (In a certain use case, see future applications)

I tested this approach using two foundation models. As the multimodal large language model (MLLM), I used the OpenAI API with GPT-5, and as the visual model, I used SAM. Both models are pretrained on a large amount of data (Bommasani et al., 2021) and therefore do not necessarily require additional fine-tuning.



Figure 1 Overview of the tested workflow. Raw Extraction shows the initially generated building polygon with an overlaid grid used for MLLM processing. MLLM QA illustrates the quality assessment step, where the multimodal language model identifies and suggests geometric improvements. Refined Result displays the final building footprint after applying the suggested refinements (in solid red).

For the final master's thesis, it still needs to be decided whether foundation models such as SAM, GPT, LLaVA 1.6 / 1.7 (Vicuna-13B), or InternVL 2 (20B) will be used, or if instead specific models will be trained for this task (e.g., LaMa v or similar architectures).

As a reference and benchmark, either the swisstopo TLM dataset together with SwissImage aerial imagery ([swisstopo, 2025](#)) or publicly available building datasets such as Microsoft Open Buildings and Google Open Buildings will be used.

The swisstopo dataset is subject to licensing agreements that restrict its use with commercial services such as the OpenAI API ([Terms of Use](#)). Therefore, the goal is to use an open-source multimodal model that can be run locally in a Docker container or on a Swiss-based server to ensure compliance with data protection and licensing requirements.

Regarding the motivation and purpose of this work:

The task of building footprint extraction originates from the geospatial industry, for example at swisstopo, where according to a short informal interview with an operator there is still no fully operational automated system capable of reliably generating building footprints for new or changed buildings.

The overall goal of this study is to explore whether combining visual and multimodal foundation models can improve the last missing 5–10 % of correctness that often remains in existing deep-learning-based extraction workflows (Adimoolam et al., 2024) and to explore its potential applicability in other remote sensing tasks (see Future Applications and Scope Expansion). By adding an additional quality-assurance layer before the final building footprint generation, the system could provide a more accurate and reliable footprint suggestion, reducing manual post-editing and improving production efficiency.

Future Applications and Scope Expansion

The project will first focus on a clearly defined application in developed regions (e.g., Switzerland), where high-quality datasets and existing benchmarks are available. Once the workflow is successfully established and validated, and **if time and capacity allow**, further

applications can be explored — such as transferability testing in underdeveloped regions, detecting building damage from extreme weather or armed conflict, or assessing urban change. This progressive approach ensures a strong methodological foundation before applying the framework to broader humanitarian or global development contexts.

Period	Phase	Key Tasks & Milestones
Nov – Dec 2025	Literature Review & Concept Development	- Read and summarize core papers (CNNs → DPT → SAM → MLLM). - Define research questions, objectives, and hypothesis. - Write Introduction draft and Theoretical framework. - Set up environment (Docker, GPU access, Swisstopo data). <i>Milestone 1: Approved proposal & first theory draft.</i>
Jan – Feb 2026	Visual Model Implementation (eg SAM)	- Implement footprint extraction workflow. - Evaluate baseline results (IoU, precision, recall). - Continue writing Methodology – Part 1 (Visual Model). <i>Milestone 2: Working baseline & documented visual workflow.</i>
Mar 2026	MLLM QA Integration	- Implement and test MLLM evaluation stage (GPT-4V, InternVL2, etc.). - Define JSON schema for issues & suggestions. - Draft Methodology – Part 2 (MLLM QA). <i>Milestone 3: Functional QA pipeline with description in text.</i>
Apr – May 2026	Model Coupling & Framework Development	- Integrate MLLM + Visual Model into single pipeline. - Automate improvement suggestions and feedback loop. - Update Methodology with diagrams and flowcharts. <i>Milestone 4: Prototype framework complete & method fully written.</i>
Jun 2026	Evaluation & Analysis	- Run large-scale experiments on dataset. - Maybe expand Application case (other countries, change detection etc) - Compare performance vs. baseline and between models. - Begin Results chapter (quantitative tables, figures). <i>Milestone 5: Evaluation dataset processed & first results written.</i>
Jul – Aug 2026	Discussion, Conclusion & Finalization	- Write Discussion, Conclusion, and Abstract. - Final proofreading, formatting, and supervisor feedback. - Prepare presentation (if required). <i>Milestone 6: Final thesis submission (mid-August).</i>

AI Statement: ChatGPT 5 was used to help generate and structure the timeline table, and to refine the wording. It also supported me in improving sentence structure, understanding, spelling, and grammar. In addition, it assisted with citation formatting.

Literature:

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